



CSC-361 – Machine Learning | Complex Computing Problem

Semester: Spring 2025

Max Marks: 7

Track: Track A – Research-Oriented Project

Mapped CLO: CLO-2

Instructor: Dr. Shafiq Ur Rehman Khan

Track A – Research-Oriented Project
Milestone 2 – Experimental Setup and Baseline Model

Course: CSC-361 Machine Learning	Semester: Spring 2025
Track: Track A – Research-Oriented Project	Due Date: 30 April 2025
Instructor: Dr. Shafiq Ur Rehman Khan	Max Marks: 7 / 20
Mapped CLO: CLO-2	

Mapped Course Learning Outcome (CLO)

- CLO-2: Apply regression, classification, and clustering algorithms to solve standard machine learning problems.

Milestone Objective

Students will implement data preprocessing, train a baseline ML model using regression, classification, or clustering algorithms, and report initial evaluation results. This milestone directly assesses the ability to apply appropriate ML algorithms to the chosen research problem.

Required Deliverables

- Written Report (PDF, 3–4 pages)
- Presentation (7–10 minutes)
- GitHub repository with training/testing code and README

Submission Requirements

- Dataset description (features, size, class distribution)
- Data preprocessing steps applied
- Baseline model selection with justification — regression / classification / clustering
- Training procedure description with train/test split
- Evaluation using appropriate metrics: accuracy, precision, recall, F1-score, MSE/RMSE
- Initial performance results with brief analysis

Evaluation Rubric — Total Marks: 7 / 20



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Group Name / No.:

Submission Date:

Criterion	Excellent (Full Marks)	Good (75%)	Satisfactory (50%)	Needs Work (25%)	Marks	CLO
Data Preprocessing	Comprehensive preprocessing: missing values, encoding, scaling/normalization all handled	Most preprocessing steps applied; minor gaps	Some preprocessing done but incomplete	Little or no preprocessing performed	1.5	CLO-2
Baseline Model Implementation	Correct, working regression/classification/clustering algorithm with clean code	Model works with minor issues	Model partially implemented	No working model	2.0	CLO-2
Evaluation Metrics Used	Multiple appropriate metrics (e.g., Accuracy + F1 + Recall, or MSE + R ²) reported clearly	At least two metrics used and reported	Only a single metric used (e.g., accuracy alone)	No metrics reported	1.5	CLO-2
Results Analysis	Clear analysis of initial results with discussion of model strengths/weaknesses	Results reported with partial analysis	Results listed but no analysis provided	No results or analysis	1.0	CLO-2
Code Quality & GitHub	Clean, commented code; GitHub link functional with clear README	Code mostly clean; README present but incomplete	Code submitted but poorly organized; no README	No code or GitHub link	0.75	CLO-2
Presentation	Confident delivery; dataset, model, and metrics clearly explained within time	Good delivery; most aspects covered	Partial coverage; some aspects missing	Poor delivery; major aspects missing	0.25	CLO-2

Marks Distribution Summary

Criterion	Max Marks	CLO
Data Preprocessing	1.5	CLO-2
Baseline Model Implementation	2.0	CLO-2
Evaluation Metrics Used	1.5	CLO-2
Results Analysis	1.0	CLO-2



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Code Quality & GitHub	0.75	CLO-2
Presentation	0.25	CLO-2
Total	7.00	

Important Notes: All group members must be able to explain their individual contribution during viva. Plagiarism in report or code will result in zero marks for the entire group. Accuracy alone is not an acceptable evaluation metric — use multiple metrics appropriate to the ML task.